Todo list

Typesetting a Thesis

Make thesis writing as
funny as it can possibly get!

Der Fakultät für Mathematik und Informatik
der Universität Leipzig
eingereichte

D I S S E R T A T I O N

zur Erlangung des akademischen Grades

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im Fachgebiet

Informatik,

vorgelegt von

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KEEP
CALM
AND
RTFM

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Abstract

In this manual, you get an introduction in typesetting a thesis with latex and especially with the UniLeipzigBioinfThesis class. The core is the use of the class. However, there are also some hints and suggestions on what you should do when you write your thesis. It starts with an introductory chapter. Followed by three chapters that cover the main parts of a thesis: front matter, main text and back matter.

It is designed to fulfill the “Promotionsordung” from the “Institut für Informatik” at the “Universität Leipzig”. The result is a class that can be used without any further tweaking and create a good looking thesis that contains all required parts. However, you can modify many of the design decisions so that you can adapt it to your flavor.

Thus, the class helps you to directly start writing and not think too much about the typesetting. I hope you enjoy this class.

Zusammenfassung

Dunkel war’s, der Mond schien helle,
schneebedeckt die grüne Flur,
als ein Wagen blitzesschnelle,
langsam um die Ecke fuhr.

Drinnen saßen stehend Leute,
schweigend ins Gespräch vertieft,
als ein totgeschoss’ner Hase
auf der Sandbank Schlittschuh lief.

Und ein blondgelockter Jüngling
mit kohlrabenschwarzem Haar
saß auf einer grünen Kiste,
die rot angestrichen war.

Neben ihm ’ne alte Schrulle,
zählte kaum erst sechzehn Jahr,
in der Hand ’ne Butterstulle,
die mit Schmalz bestrichen war.

Acknowledgments

I thank everybody who helped me to create this class and correct some of my more serious mistakes. I also thank my family who makes it possible that I work on this besides my thesis work. After all, I have not enough time for them.

Furthermore, I thank my fellow students and my dissertation adviser for some clues to what should be in included in this class. Finally, I thank you for looking in this manual so that it has not been done in vain.

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Introduction

CHAPTER 1

Introduction

Contents

1.1 Welcome

Welcome to the manual of the Uni Leipzig Bioinf Thesis class. This class gives a solid backbone for your thesis writing at the “Institut für Informatik” at the “Universität Leipzig”. It can be used as it is without any further configuration. However, many things can be configured. So hopefully it fits your needs and your idea of design.

This template is designed to fulfill the requirements of the “Promotionsordnung Fakultät für Mathematik und Informatik” as it is in 2018 (Fakultät für Mathematik und Informatik, 2017; Mathematik und Informatik, 2013). So in case of changes in the regulation some parts of this work may not longer be suitable.

This class has many class options and also introduces some new commands that can be used. So when you really want to use it you should at least read Appendix A and Appendix B. However, I would recommend you to read the complete manual so you get an overview how things work and how things should be done. Also, the short overviews in the Appendix reflect not every single aspect of the commands and options.

This manual is not an introduction to LaTeX. The authors assume that you already typeset a scientific work with LaTeX before and that you do not need an introduction to the basic commands. The typesetting of mathematical formulas is not covered, either.

The use of this class is free and you can change things by yourself and redistribute it if you like. However, if you use it and it makes your life easier, I would appreciate it when you mention me in your acknowledgments. Something short like: “Thanks to Fabian Gärtner for the latex class of this thesis.”

If something does not work as expected, something needs an update because of some changes or you miss an option to modify the template feel free to write me an email: `fabian@bioinf.uni-leipzig.de`. I will see if I can help you.

1.1.1 Disclaimer

Before we start, here is a little warning. At the time I made this template, I was a PhD student. I have tried to be as accurate as possible in everything that is written in this manual, and also in choosing its default options. However, some people might disagree with me and there is not just one correct answer but rather several possible options. So trust your supervisor more than anything I have done here. Again if you get some feedback that something has to change I would appreciate hearing from you such that I can include changes in order to keep the template updated.

1.1.2 This Manual

This manual was itself created with the given `UniLeipzigBioinfThesis` class. In the original download, the source files contain the code of this manual so that you have direct examples how things are done in this class. Of course, you will delete it in order to write your own content in the files. However, before you do it you may have a look at the content and keep the pdf file in a separate location such that you can reread the manual. You can also download it again in order to see the original sources.

The only difference between this source and the source of `manual.pdf` is the class option `minted` (see Section 3.10.2) and `zusammenpath` (see Section 2.5) are set different. This is done to make it compile and a offer good starting point without any further changes.

1.2 Double-sided Print

This style only supports double-sided layout for the thesis. This is not a lack of an option, this is done purposely. When reading on a screen, it makes more or less no difference if the document is single-sided or double-sided. The difference is in the way it is printed. A double-sided print is done such that every sheet of paper is printed on both sides. This has only advantages. It is closer to normal books and therefore more natural to read. It makes the thesis thinner so that it is easier to handle. It saves you money because you print on and tie less pages. The last and best reason is that it reduces the waste of paper and makes the world a better place. I believe this is the reason why we all write a thesis, to make the world a better place.

1.3 File Structure

This template splits the content of the thesis into various different files. This helps you to keep your work organized and allows you to only compile the parts that you currently edit. It also enables the automatic generation of stand-alone documents from several parts of the work (e. g. the abstract), which are formally required for the Dekanat when you hand in your thesis. Other files provide the functionality of the class itself and need not be changed. Another set of files helps you to compile the thesis and store your personal information.

The thesis itself is compiled from the file `main.tex` in the root directory. In this file, the thesis class is loaded and basic configuration is done. Especially, every path to another file can be changed here so that all files can be moved or renamed if necessary. More details are given in Section 1.5. Then, all the content from the subfolders is included and compiled into a single document. The class itself is taking care of creating the front matter and back matter for you. In addition to the main file, one tex file for each stand-alone document created from the thesis content is provided:

`abstract_sa.tex`: The stand-alone abstract.

`curriculum_scientiae_sa.tex`: The curriculum outlining your scientific career.

`publications_sa.tex`: The list of all your scientific publications.

`selbststaendigkeitserklaerung_sa.tex`: The declaration that you are the original author of all your thesis' content and that you have properly marked all citations.

The classes for the thesis and the stand-alone documents are defined in the files `UniLeipzigBioinfThesis.cls` and `UniLeipzigBioinfThesisSA.cls`, respectively. A style file, `UniLeipzigBioinfThesisCommon.sty`, contains code that is shared by both classes. These classes make all commands and styles available to the tex files. You should not edit these files unless you want to add a new feature or have to fix a bug.

The files `Makefile` and `latexmkrc` help you to compile the documents, as explained in Section 1.4.

The folders have thematically different content. The `frontmatter` folder contains files that are used in the front matter (abstract, acknowledgments,...), the `chapters` folder contains a single tex file for each chapter of the main text. The `graphics` folder contains the graphics for the figures of the thesis. The `appendices` folder contains a single tex file for each appendix. The last folder is `backmatter` which contains files that are used in the back matter (e. g. the Curriculum Scientiae). The `misc` folder contains two data files and a tex file. `literature.bib` is the *BibTeX* file that contains all works that you cited from. Your personal data and thesis information like title, keywords etc., is stored in `thesis_info.tex`. File `custom_defs.tex` is for all commands you may want to define yourself, as well as for loading other packages you may want to use. It comes with some commands already that you can use or change if you like.

Finally, `manual.pdf` contains a copy of this manual. You may want to keep it for future reference.

1.4 How to Compile It

The short version: run `make` to compile everything, or `make main` to only compile the thesis. Run `make watch` to continuously compile the thesis as you write. If you have problems or want more information, please read on.

As of release v2.0 of this class, the build system has been completely rewritten. It now relies on *latexmk*, a dedicated tool for compiling LaTeX documents that is taking care of all required steps, including the execution of *bibtex*, *biber*, *makeindex*, *pdflatex* etc., in the right order and as often as necessary. It is also more efficient than manually calling all these tools, because it determines which files have changed and regenerates the output as necessary. Furthermore, it offers file monitoring to automatically recompile the document as soon as any changes to the source code are saved, as well as cleaning procedures. Still, a `Makefile` is provided to allow calling *latexmk* with a single command without having to remember all the options and paths. For more information on how to compile using the provided `Makefile`, refer to Section 1.4.2.

1.4.1 Configuring *latexmk*

With this template comes the file `latexmkrc`, which already contains the required configuration for *latexmk*. Separating the configuration from the `Makefile` has the advantage that *latexmk* can also be called via any third-party LaTeX development environment or editor without further configuration. For example, the compilation feature of the *vimtex* plugin for the editor *vim* works out of the box with the provided files.

1.4.2 Makefile

The provided `Makefile` is divided into the sections “Config” and “Rules”. Usually, you don’t have to touch the rules at all, and changing the configuration variables is only necessary when you rename the main or stand-alone tex files.

To compile the thesis and all other stand-alone document, run `make all` or simply `make` in the thesis directory. `all` is called the (default) *target*, i. e. the object that is to be made. The `Makefile` also defines several other useful targets:

main: Compiles the main file, i. e. the thesis. The main file is by definition the first file listed in the variable `TEX_FILES`. Running `make main` is the quickest way to compile only the thesis, omitting the stand-alone documents.

***filename*.pdf:** Compiles (only) `filename.pdf` from source `filename.tex` for any file name.

clean: Removes *all* generated files, including the pdf files.

watch: Watches for changes to any of the sources of the main file and recompiles them as soon as any are detected.

watch-filename.tex: Like `watch`, but watch `filename.tex` instead of the main file.

You can also set the path to *latexmk* and additional command line arguments in the `Makefile`. A few additional targets are provided for development purposes:

dist: Make a tarball (`.tar.gz` archive) of the thesis directory. Target `clean` is executed first. The file name can be set by changing the value of variable `DIST_NAME`. Set `DIST_EXCLUDE` to prevent certain files or directories from being included in the archive.

changes: Update `CHANGES.md` by converting `appendices/changes.tex` (i. e. the version history) to MarkDown using *Pandoc*.

html: Generate HTML versions of `README.md` and `appendices/changes.tex` to update the website of the thesis template.[1]

1.4.3 Dependencies and Compiling at the Bioinf Institute

To compile this template, you obviously need a LaTeX distribution. Under Linux, this is usually *TeXLive*, which is available in the official package repositories of any major Linux distribution. However, as the template has many features, it also depends on many packages. The easiest way to provide them all is a full installation of all *TeXLive* packages. In Fedora 36, this is done by installing package `texlive-scheme-full` using `dnf`. In Ubuntu 22.04, install `texlive-full` and `python3-pygments` with `apt`. Also make sure that `make` is available if you want to use the `Makefile`. As of 2022, the computers at the Bioinformatics institute run varying versions of Fedora. If the required packages are not installed on your machine yet, please send a short email to the system administrator — Jens will gladly help you out.

Though this template has not been tested on Windows, you may have luck with *MikTeX* as well.

[1] `https://www.bioinf.uni-leipzig.de/intern/phd-thesis-class/`

1.5 The Main File

The main file can be divided into two parts: the preamble and the document body. In the preamble the class is loaded with the desired options. In the document body, the actual content of the thesis is included from other tex files.

1.5.1 Preamble

The first thing that is done in the preamble is loading the required classes. This is done with `\documentclass[]{UniLeipzigBioinfThesis}`. Class options can be added in the square brackets. The possible options are spread in the manual but all are marked with a margin note.

After this, you need to attach one or more bib files to you thesis this is explained in Section 4.3. You can also defined additional commands and load other packages here if necessary.

1.5.2 Document Body

The document starts with the `\makefrontmatter` command. This command creates the complete front matter of the thesis. This includes the title page, the dedication, the bibliographic description, the abstract, the German abstract, the acknowledgments and Table of Contents (ToC). Make sure that your personal data in file `thesis_info.tex` is correct as it is used to generate the front matter.

The main matter of the thesis start with the call of the `\beginmainmatter` command. The command starts the page numbering with Arabic numerals. The main matter is composed of chapters. Every chapter is in a separated file. This files then are included in the main file with the `\include` command. For more details on chapters, have a look at Section 3.2. The only other elements in the main matter are the parts. Parts are described in more detail in Section 3.1.

The next part is the appendix. To start this part use the `\beginappendix` command. It changes the chapter behavior and adds an entry to the ToC. Every appendix is in an extra file. To add these files, use the `\include` command again. For more information, see Section 4.1. The last command is then `\makebackmatter`. This command creates the back matter that includes the List of Xs, bibliography, curriculum scientiae and the declaration.

1.6 Base Settings

1.6.1 Margins

By default, the template uses relatively wide margins as defined by the default settings of the *Memoir* class. Though this may seem unusual first, especially when you are used to a typical Word document, but it is generally considered the more aesthetic choice by most typographers. It also reduces the eye movement compared to longer lines and thus can be read more easily. However, when you have much to say, this can make your thesis very large and heavy when printing. Thus, it could make sense to reduce the margin size. This of course reduces the space for margin notes and I would not use them with small margins. You can

smallmargins change to smaller margins with the class option `smallmargins`.

Value	Font used
lm	Latin modern without serifs
lms	Latin modern with serifs (default)
cm	Computer modern without serifs
cms	Computer modern with serifs

Table 1: Font values. The values that the class option “font” can have.

1.6.2 Font

The default font that is used in the tex files is Latin modern with serifs. This is different to usual latex documents that use Computer modern with serifs as default font, but the result is usually much better. You can change the font with the class option `font`. The possibilities are shown in Table 1.

font

1.6.3 Chapter Design

There are many different possibilities for chapter page design. To change this, use the class option `chapterstyle`.. An example of the possible designs is shown in Figures 1–6. However, you could use any design of the *Memoir* class shown in (Wilson, 2018, P:383-396). The default value is the *veelo* style, which has a mini ToC within each start page of a new chapter.

chapterstyle

1.7 Common Typographic Mistakes

While LaTeX is designed to generate documents with a high typographic quality, there are a few common pitfalls that authors have to be aware of to avoid unexpected or unpleasant results. Here, some of the most relevant tricky details to keep in mind are described.

1.7.1 Logical Markup

One of the most important rules to achieve a consistent typography in your LaTeX documents is to consequently use *logical markup* like `\emph` (for emphasis) or `\email` (for an email address) instead of directly styling the text with low-level directives like, e. g., `\textit` (for italic text) or `\tiny` (for small a font size). Logical markup means that you apply a command to a piece of text that describes *what it is*, and not what it looks like. For example, a file name is often typeset in typewriter font. However, instead of directly using `\texttt{the_name}` to achieve this, it is better to define a command `\file{}` for this purpose and use `\file{the_name}`. This has multiple benefits: (1) All file names using this command will look identically, and you do not need to remember the exact style. (2) You can easily change the style later by simply changing definition of the logical markup command. (3) A custom command may be shorter than the basic LaTeX commands, especially if you have to use multiple ones to achieve the desired look. (4) You can simply find all file names etc. in the document by searching for the respective logical markup command. In case you want to get rid of a logical markup command later, you can usually simply replace all occurrences using a regular expression substitution.

To define your logical markup commands, use the `\newcommand` command. User-defined commands should be defined in file `misc/custom_defs.tex`. It

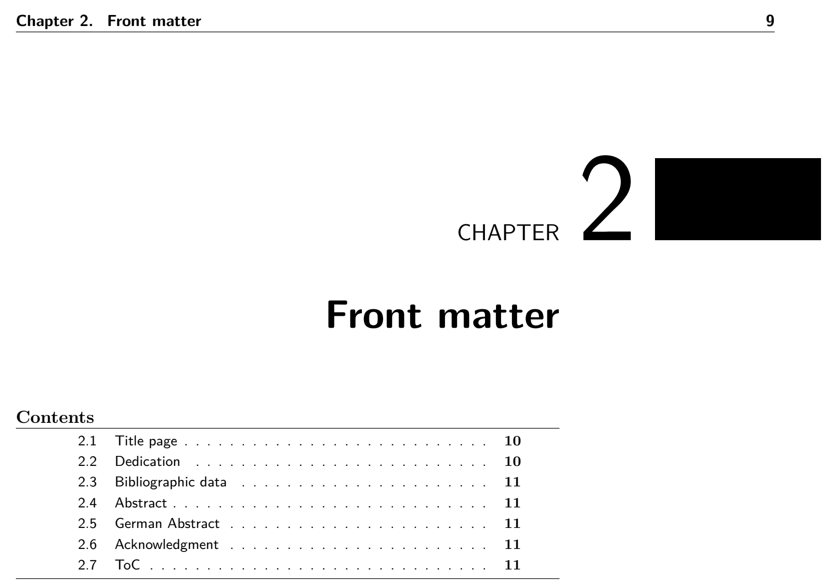

Figure 1: Veelo page style example. An example how the veelo page style look with a mini ToC

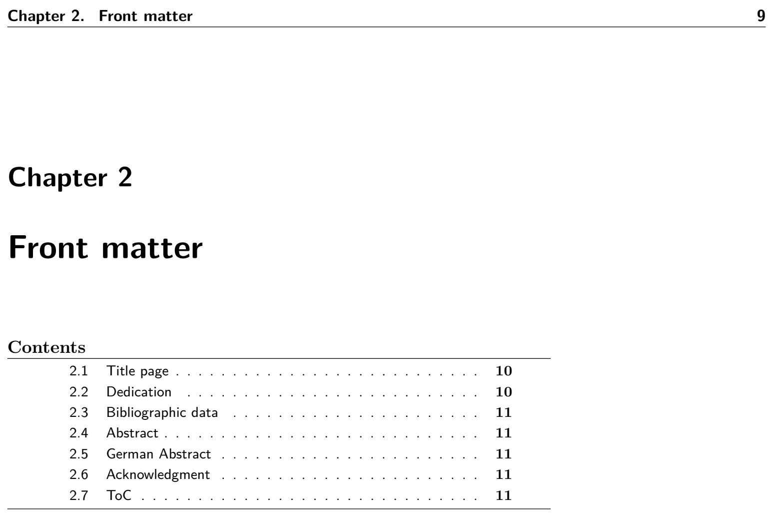

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Figure 2: Default page style example. An example how the default page style look with a mini ToC

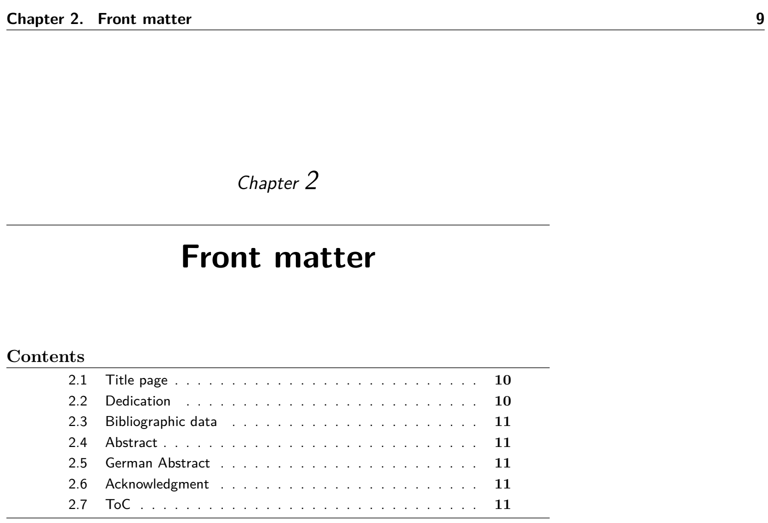

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Figure 3: Bianchi page style example. An example how the bianchi page style look with a mini ToC

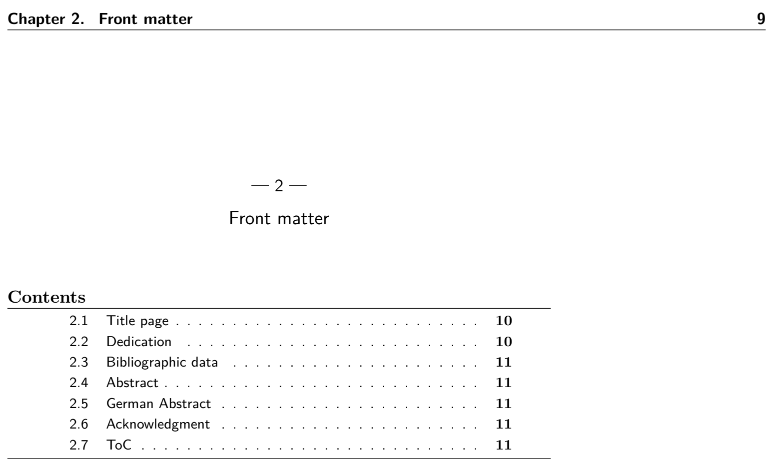

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Figure 4: Dash page style example. An example how the dash page style look with a mini ToC

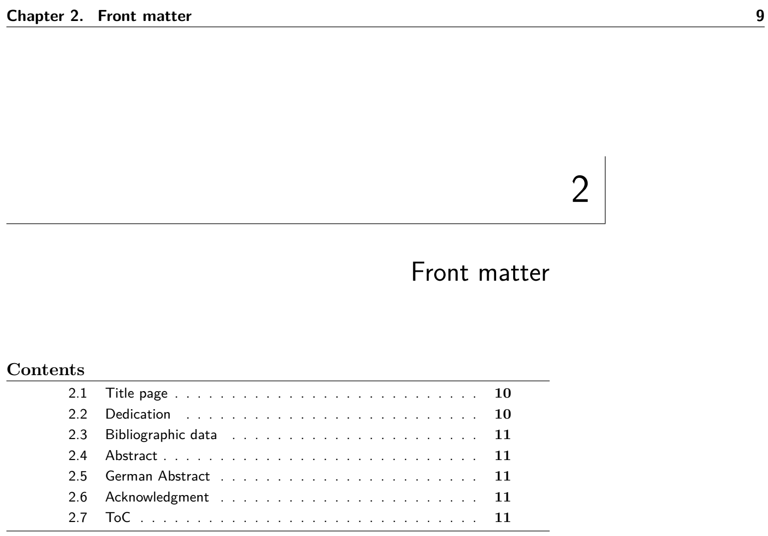
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Figure 5: Ell page style example. An example how the ell page style look with a mini ToC

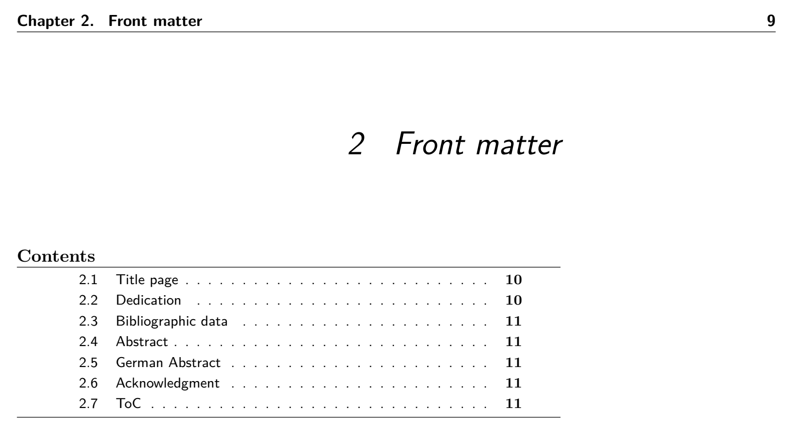

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Figure 6: Wilsondob page style example. An example how the wilsondob page style look with a mini ToC

Command	Description
`\species`	species name
`\seq`	DNA/RNA sequence
`\file`	file name
`\cmd`	code or command
`\prog`	program name
`\email`	email address
`\quo`	quotation marks
`\tblhead`	table headers
`\pkg`	software package/module

Table 2: List of the default logical markup commands, which are defined in `misc/custom_defs.tex`.

already contains pre-defined commands for the most common text elements in our fields, e. g. `\species` for species names, `\prog` for computer program names, and `\cmd` for shell commands and code. For a complete list, refer to Table 2. Of course, you can extend the list or change the styles if you like.

1.7.2 Spaces

Inter-word spaces are a common source of typographic mistakes. This is mainly because, in LaTeX, not every space has the same size. Most notably, it increases the space after a period ("."), which is usually good as it helps to visually separate different sentences. However, the period is also used to mark abbreviations, which leads to wrong spacing as, e. g., in Mr. A. Brown. The solution is to escape the following space, as in Mr. A. Brown (using the code `Mr.\ A.\ Brown`).

When defining a command which ends in a period, you do not know whether there will be a space after it or not. In such cases, the sequence `\@` can be used to disable the increased size of a space that will possibly follow. For example, `\newcommand{\dr}{Dr.\@}` defines the command `\dr`, which is then displayed as Dr. with correct spacing.

Beside regular spaces, LaTeX also offers the command `\,` to generate half spaces. They are useful to group digits like 1 500 000 or to typeset abbreviations consisting of multiple words like w. r. t. ("with respect to"), which looks much better than both w. r. t. (using full spaces) or w.r.t. (using no spaces).

Another variety of the normal space is the *unbreakable* space, written as tilde character (`~`). It has the same size as a regular space, but prevents the line from being broken at this specific point. This is useful for groups of text elements that belong together and should not be separated, for example a title and a name like Mr. Brown, or to prevent that abbreviations followed by a period end a line and thus confuse the reader as it looks as though the period ends the sentence.

A final recommendation is related to the somewhat surprising behaviour of LaTeX when it comes to spaces following commands. Using a command without any arguments like `\dr` defined above "swallows" the following space, i. e. it will print Dr.and no space after it like so. To fix this, either use `\dr{}`, or use the `\xspace` command from the `xspace` package in the definition: `\newcommand{\dr}{Dr.\@\xspace}`. It will expand to a space only when needed and not, e. g., when used within parentheses. The pre-defined markup commands in `misc/custom_defs.tex` showcase the use of `\xspace`.

1.7.3 Quotation Marks

In short: always use the `\quo{}` command to quote your text like “this”.

While using proper quotation marks like “these” seems to be easy at a first glance, it turns out to be a complex matter. First, one should realize that the double quote character `"` looks horrid and should never be used in a LaTeX text. Instead, two backticks (` `` `) and two apostrophes `''` are used to create correct double quotation marks as in the first sentence. Note how the opening and closing marks differ slightly. To make things worse, quotation marks are language-specific, so in the German Zusammenfassung, you would for instance use „diese Art von“ quotation marks. Since this is easy to get wrong, this thesis class properly sets up the language settings for the respective parts and provides the command `\enquote{text}` from the package `csquotes`, as well as the shorthand `\quo{text}`. Using this command is not only convenient and guaranteed to produce correct results, it also handles edge cases like inner and outer quotes: “‘Hello,’ he said.”

1.7.4 Dashes

Texts contain a wide variety of dashes—i. e. horizontal lines—that have different lengths depending on their function. To insert partial sentences that could also be surrounded by commas or parentheses, or to append them as if following a semicolon, the *em-dash* (German Geviertstrich) is used. It is named after the widest letter M. In LaTeX, it is typeset using three dash characters (`---`). The first sentence of this section demonstrates its usage. The shorter *en-dash* (German Halbgeviertstrich), has half the size of an em-dash. It can be typeset using two dashes (`--`). It is used to indicate ranges (e. g. 10–15 minutes), or for compound nouns as in Watson–Crick base pair or protein–protein interaction. The shortest dash is the hyphen (German Viertelgeviertstrich), written simply as `-` in the code. It is used to hyphenate words at the end of lines, or for compound adjectives like “fast-growing” and sometimes for prefixes like in “re-evaluate”.

To get correctly sized minus signs, always set negative numbers in a math environment: -5 (code: `$-5$`) and not -5.

1.7.5 Position Specifiers of Floats

Do not overuse position specifiers of floats environments. For example, the figure environment allows to pass the specifiers h (*here*), t (*top*), and b (*bottom*) as in `\begin{figure}[htb]`. Do not use them unless you have a good reason. Do not add them to all of your figures by default. LaTeX usually does a good job finding a suitable position on its own. If you feel you have to add specifiers, do it when you are close to finishing that chapter, when you have already included all other figures, or you may get surprising results.

1.7.6 Tables

Instead of using `\hrule` to draw horizontal lines in tables, use `\toprule`, `\midrule`, and `\bottomrule` as in the tables of this manual, e. g. Table 2. Do not overuse these commands as too many lines will only clutter your table, but not benefit its readability.

Part II

Content

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Front Matter

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Table 3: Titlepage commands The commands that are used in the preamble of `main.tex` to input the information of the title page. Also is given if the information is used in the submission (sub) version or the bib version.

Command	Description	Used for
`\settitle`	The title of the thesis.	sub, bib
`\setsubtitle`	The subtitle of the thesis.	sub, bib
`\setauthortitle`	The academic degree if the author.	sub, bib
`\setauthor`	The full name of the author.	sub, bib
`\setdayofbirth`	The day of birth of the author.	sub, bib
`\setbirthplace`	The birth place of the author.	sub, bib
`\setfilingdate`	The date where you submit your thesis.	sub
`\setfirstreviewer`	The first reviewer of you thesis	bib
`\setsecondreviewer`	The second reviewer of you thesis	bib
`\setdayofdefens`	The day where you defended your thesis	bib
`\setgrade`	The grade that you get on your thesis	bib

2.1 Title Page

The first page of the work is the title page. The design of this page is specified by Mathematik und Informatik (2013). There exist two versions of the title page for different applications. The first one is the title page for your submission of the thesis. After a successful defense of your thesis, it needs to be published. This is done by uploading your work to `http://ul.qucosa.de`. However, this version needs a different title page. This is the second version of the title page. To create this title page instead of the submission title page use the class option `bibversion`.

bibversion

The information on this page is provided by some commands in the data file `thesis_info.tex`. An overview of this commands is shown in Table 3. Note that most of the information exclusive for the bib version can only be given after you defense. This is no problem because you only need this version afterwards.

A little note to the `\setsubtitle`: Not every work has or needs a subtitle. In this case, you can simply not use the command anywhere and the class eliminates the appearance of the subtitle everywhere.

2.2 Dedication

The next page after the title page is the Dedication page. Here you can dedicate your work to someone or make a suitable insider joke. Whatever it is, it should be short and in the best case one sentence. You can input this sentence with the `\setdedication` command in the preamble of `main.tex`.

If you do not want to dedicate your work to someone then leave the command empty (`\setdedication{}`). This eliminates the dedication page. Note that the content of `\setdedication` can be any latex command so it not only possible to write text it could also include a picture or a combination of both.

2.3 Bibliographic Data

The third nonempty page contains the bibliographic data of the thesis. This is required by Mathematik und Informatik (2013). Most of the data is calculated automatically. The only exceptions are the keywords of the thesis which are inserted by the command `\setkeywords` in the preamble of `main.tex`. The keyword should be given as a comma-separated list.

2.3.1 Statistics

In the middle of the page are some automatically calculated value. These values are the core of the bibliographic data. However, they are not mandatory from the Promotionsordnung (Mathematik und Informatik, 2013). Thus, you can deactivate them with the class option `nostatistics`. Then nothing is calculated or shown, and only the keywords and the paper the thesis is based on are displayed. **nostatistics**

2.3.2 Thesis Publication

Most likely you have published already some of the results you present in this thesis. Such a publication is then a publication on which the thesis is based on. They can be divided from publications you only cited by the fact that in some way you retell them in the thesis. This can even mean that long paragraphs are equal between them and the text of the thesis. This information is critical to prevent the accusation of plagiarism. So this publications belongs to the bibliographic data of the thesis.

To add publications there is the command `\setbase` in the preamble used. The command gets a comma-separated list of BibTeX keys. Then it automatically fetches the full information out of your BibTeX file. However, this, of course, means that this publication must be in the BibTeX file of the thesis.

The look of this paper can be influenced with the class option `baseonauthorstyle`. **baseonauthorstyle** This change how the authors are displayed. More details can be found at Section 4.3.2. The default value is “gf” which means that first the given name comes and then the family name.

A second possibility to change the look of the author respectively give extra information of the authors. This is done by annotations. A deeper look into annotations can be found in Section 4.3.3. Two things are vary handy here. First to mark authors that share a first authorship. This make it possible to mark that you are a first author even when you are not on first position. The second thing is the possibility to underline specific authors. This is useful when you want to mark you in the papers. Maybe when you not on first position or when you have changes your name by a marriage. Even when both is not given a mark is a good idea to make the reader easy to spot you.

2.4 Abstract

Then the abstract of the work follows. This is limited to four pages (Mathematik und Informatik, 2013). The abstract is written in an extra file. By default this file is placed at `frontmatter/abstract.tex`. You can change the path to the

abstractpath abstract with the class option `abstractpath`. Note that you have to *omit* the `.tex` extension when defining an alternative value for this option.

A stand-alone version of the abstract is generated automatically. You are required to hand it in together with your thesis. It will be handed out during your defense.

2.5 German Abstract

The template gives the option to include a German version of your abstract. This is mandatory when you write a binational thesis. If you only write it for the University of Leipzig you can choose if you want to include it. However, most people add a German abstract.

The default path is `frontmatter/zusammenfassung` (omit the `.tex` extension). Again you can change the path to the file that contains the German abstract with the class option `zusammenpath`. If you do not want the German abstract, you can simply set the class option to an empty value (`zusammenpath=,`).

zusammenpath

2.6 Acknowledgments

The third part of the front matter that inputs text from another path is the acknowledgments. Here you should thank everybody that helped you with the thesis: family, friends, supervisor, fellow students and maybe me.

acknowledgmentspath The class option `acknowledgmentspath` changes the path to the file. The default path is `frontmatter/acknowledgments` (omit the `.tex` extension).

2.7 The Table of Contents

The Table of Contents (ToC) is directly in front of the main matter. It only contains entries from the main matter and the back matter. The ToC shows entries down to the section level. Subsections are not shown here. However, they are shown in mini ToCs at the beginning of each chapter. More details on mini ToCs are given in Section 3.2.

CHAPTER 3

Main Matter

Contents

3.1 Parts

The highest form of document division is *part*. This division is not required at all. However, you can use it to give you work some structure. There should not be more than three or four parts in the complete thesis. Typical cases would be:

- I Background or Introduction
- II Methodology
- III Application

To start a new part the command `\part{title}` is used. This creates a part page where the title is shown. It also generates an entry in the ToC. The parts are automatically numbered with roman numbers but have no influence on other numberings.

3.2 Chapters

The most important document division is the chapter. A new chapter is started with `\chapter{title}`. This creates a chapter page and generates an entry in the ToC. How the chapter page is designed as described in Section 1.6.3. The chapters are numbered with arabic numbers. All following sections and subsections are in the chapter and thus, have the chapter number. The header also contains the chapter information. Thus a new chapter changes the look of the header.

Maybe at different locations, the name of the chapter should be different. There are three positions: First the entry in the ToC. Second, the entry in the header of the page and third is the title on the chapter page. It is possible to give for each of the three locations a different title. Different title version can be useful as an example when the title is too long for the header of the page. For this the `\chapter` command has two optional arguments. The command `\chapter[TOC][HEADER]{TITLE}` generates in the ToC the entry “TOC”. In the header, it generates the entry “HEADER” and on the chapter page the entry “TITLE”.

3.2.1 Mini TOC

At the beginning of every chapter in the main matter, a mini ToC is printed. This contains all sections and subsections of the chapter. The advantage here is that the reader gets an overview directly at the beginning of each chapter and doesn’t have turn back to look at the main ToC.

nominitoc

However, this needs some space and some people say it is page flay. So you may want to disable the mini ToC. To do this, set the class option `nominitoc`. When the mini ToC is disabled the text of the chapter begins directly on the same page under the chapter title.

3.3 Labels and References

For internal references, the `\label{key}` and the `\ref{label}` commands are normally used. This is fine and also works for this class. However, there is a

Algorithm 1: Referencing alternatives The two referencing methods that create the same output.

```
See Figure~\ref{fig:veelo}.
See \autoref{fig:veelo}.
```

second option: the `\autoref{label}` command. The difference is that `\ref` only returns the number of the referenced object, where `\autoref` also returns a suitable prefix depending on its type (e. g. figure, section etc.). An example is shown in Algorithm 1.

The upside is that the generated hyperlinks already start at the prefix, so they can be clicked more easily. It also reduces the possibility of referencing the wrong object. In rare cases, it makes it easier to change the type of a float or a section (from/to subsection) because you do not need to change the type prefix for every reference.

The downside is that LaTeX needs to know to every type and how it should be named. Also, referencing more than one object is possibly not useful. Every element that is predefined by this class can be used here. As long you do not introduce new floats or theorems, `\autoref` is the simplest way to go. In the manual, `\ref` is only used to reference more than one object (see Section 1.6.3).

A question naturally arising when using `\autoref` is: "How is the type prefix displayed?" Should the first letter be uppercase or lowercase? The default in this template is that every type prefix is spelled with an uppercase first letter. However, this behavior can be changed with the class option `smallautoref`. When it is specified, all type prefixes are written spelled with lowercase letters. **smallautoref**

3.4 Citations

There are two main citation commands that you should use: `\citet` and `\citep`. `\citet` is used when the citation is read as part of the **t**ext (i. e. when using the author name as a noun), e. g. when saying that the `biblatex` package was written by Lehman et al. (2018). `\citep` is used when the entire citation is in **p**arentheses (which should be the default case), e. g. when simply saying that we use the `biblatex` package (Lehman et al., 2018).

Depending on whether the `bibtex` option is used for loading the class, citations are provided by the `natbib` package based on `bibtex` (`bibtex` given), or with `biblatex` in the `natbib` compatibility mode (`bibtex` *not* given). Since `biblatex` is more modern and robust, you should not specify the `bibtex` option unless you have a good reason to do so. By default, the `author-year` style is used. Thus, not a number but the author name and year is printed. This is more informative than only giving a number that you have to remember and look up in the list of references to know the author. Though numbered citations are more common in papers, you should consider that your thesis is going to be a much bigger document and the bibliography may become huge.

3.5 Margin Notes

There is the possibility to print notes in the margin of the space. More precisely on the outside margin of the page. These notes are very prominent and seen easily but destroy the reading flow because they are outside of the normal reading area. This can be used for information that should be easily found but should not be used too often because then it loses its importance and only disturbs the reading flow.

To do such a margin note use the command `\marginnote{text}`. An example is in this manual the class option. Every time the class options are presented in different sections all over the manual, the command appears as a margin note. Thus, every time a new class option is introduced a margin note gives a hint that it is presented here. This makes it possible to scan quickly for a specific class option.

3.6 Definition Index

One of the most popular uses of margin notes is to give a reader the position of a definition. Then when a term is used the reader can easily find it and check what it means. This concept can be extended by creating an index at the end of the thesis where for every definition a site is given where to find it. Such that from anywhere in a large thesis fast the right position of a definition can be found.

This can be archived with a special command. The template gives the `\indexdef{name}` command. This command creates a margin note with the given name at the side. Further when the `definitionindex` class option is given an entry in the Definition Index is created.

definitionindex

However, not every time you want to create an index you also want to create a margin note. Thus, there are two different commands also given in to create index entries. First the `\indexnomn{name}` command. This command creates an entry in the index but does not print anything at the position. It could be combined with the `\marginnote{text}` command to create different content in the margin note and the index.

The second other command is `\indexbf{name}`. This command also creates an index entry and print “name” in bold letters. Thus, a highlighting of the position is possible without the disturbing character of margin notes. It is particularly useful when small margins are used.

3.7 Abbreviations

Every thesis uses abbreviations. An abbreviation can be a generally used term like time zones (Coordinated Universal Time (UTC)) or a topic-specific thing (Directed Acyclic Graph (DAG)) but also very specific problem descriptions (minimum Feedback Arc Set (mFAS) Problem). So it is necessary to keep track of these abbreviations in a List of Abbreviations. To add a new abbreviation use the `\defabb{short}{long}` command.

Besides this list, other things must be considered. If an abbreviation is first introduced a long version must be given to explain the abbreviation. In the following, only the short version is used. However, after some pages of not using

Algorithm 2: The abbreviations engine. An example definition of Abbreviation and the use of it. It is also an entry to the List of Abbreviations created.

```
\defabb{DAG}{Directed Acyclic Graph}
The life is a \abb{DAG}.
%The life is a Directed Acyclic Graph (DAG).
Many \abbs{DAG} together create a \abb{DAG}.
%Many DAGs together create a DAG.
\defabb{SV}[SVs]{Super Vertex}[Super Vertices]
\abbs{SV}
%Super Vertices (SVs)
```

the abbreviation, it is useful to use the long version again so that the reader gets a reminder what this abbreviation stands for. Another thing that must be considered is that sometimes the plural (DAGs) of an abbreviation is used (per default an "s" is added to the end).

So the abbreviation engine gives more than the "List of Abbreviations". It also gives the possibility to access earlier defined abbreviations with the `\abb` (for plural `\abbs`) command. This returns the long version with the short version in parentheses after the long version was not used on the pages before. An example of the syntax can be found in Algorithm 2.

The default plural form with simply appending an "s" is not always the correct way to go (Super Vertices (SVs)). For such situations, the alternate command form `\defabb[label]{short}[plural short]{long}[plural long]` with the optional arguments `plural short` and `plural long` can be used. One is the short version in plural and the other the long version in the plural. Note that it is possible to give only one of the two plural forms, the other is then simply created by adding "s". The `label` argument allows to assign a label to the abbreviation such that it can be referred to using `\abb{label}`, which then still prints the supplied short or long form. With the two-argument command, the short form of the abbreviation is itself used as the label. However, for complex abbreviations including formatting and other commands, this is not possible.

The default value of pages between two uses of `\abb` before the long version is printed again is 3. This can be changed with the `maxdis` class option. A value of 0 means that only the first appearance is printed with the long version all other occurrences have the short version. **maxdis**

The functionality described here comes from the glossary package. It is a large package. Thus, it makes sense not load it when you do not want any "List of Abbreviation" at all. This deactivation is done with the class option `nostatistics`. After the deactivation, the abbreviation commands still exists but do different things. The `\defabb` command does nothing, the `\abb` command simple print the abbreviation, and the `\abbs` command print the abbreviation with an "s" at the end. This behavior makes it possible to use them and then later activate the glossary package again and have the full functionality with the commands. **nostatistics**

Algorithm 3: The sym command. An example definition of Symbol. The result can be seen in the List of Symbols.

```
\sym{C}{The circumference of a cycle.}
\sym{\pi}[kreis]{Ratio of a circle's
    circumference to its diameter.}
We use the new cool symbol in math
$\symkreis = 3.14\dots$ and in text \symkreis.

\operatorsym{lca}[lca]{The last common ancestor.}
Let \symlca be the last common ancestor. When $v$ has
two children $u$ and $w$ then is $\symlca(u,w)=v$.
```

3.8 Symbols

In the back matter, a List of Symbols is presented. To make this possible, symbols need to be defined first. To introduce a new symbol, use the command `\sym{symbol}{Description}`. The symbol can then be used in a mathematical environments as well as in text. It is also added to the List of Symbols using the provided description. An example is shown in Algorithm 3. While symbols may be defined within the document body, e. g. directly where they are used for the first time, it may be more manageable to maintain all definitions in `misc/custom_defs.tex` to have them all in a single place.

This is some extra work for the “List of Symbols”. However, this list really helps the reader to understand complex formal definitions. It is also very helpful for the writer. The list makes it easy to determine if a symbol is used twice with varying meaning. This should be avoided, even in such a complex work as a thesis.

The syntax is simple. The definition has an optional field in the middle that defines a name: `\sym{symbol}[name]{Description}`. This name can then be used to reference back to the symbol using the command `\sym*name*`. If the name is omitted, the symbol itself is used as the name. This works only if the symbol consists of regular letters, otherwise a name must be provided.

Sometimes you may want to define a custom math operator as a symbol. This can be done by using the `\DeclareMathOperator{\lca}{lca}` command in `misc/custom_defs.tex` (or another place in the preamble). Then, you can create a symbol using this operator. However, if you really want, you can also define an operator symbol in the document body with the help of the `\operatorsym{symbol}[name]{Description}` command. Note that this makes it necessary to use the `\sym*name*` command to refer to the operator. If you do not wish to have the “sym” prefix, you need to declare the math operator using `\DeclareMathOperator{\lca}{lca}`.

The symbol engine has a problem with the “|” symbol. However, this can be fixed by not using direct “|” but one of the two command replacements: `\vert` or `\mid`. The first one is more or less the same as “|” and is used for example to get the absolute value of a number. The second have more space around it and is used in constructive set definitions.

Algorithm 4: The symbol categories. An example definition of Symbol categories. The result can be seen in the List of Symbols.

```latex
%Categories can only defined in the preamble.
\addSymbolCategory{C}{Cycle}
\addSymbolCategory{T}{Tree}

\begin{document}
% Symbols can defined in the document body.
\operatorsym{lca}[lca]{The last common ancestor.}[T]
\sym{\pi}[kreis]{The ratio of a circle's
    circumference to its diameter.}[C]
\sym{C}{The circumference of a cycle.}[C]
\end{document}

```

3.8.1 Ordering and Categories

The ordering of the symbols fellow normal the ASCII code (+<1<:<A<\<a). Note symbols define with `\operatorsym` are placed with `\operator` at the beginning. Especially the fact that “\” is placed between “A-Z” and “a-z” in the ordering is often some way counterintuitive. To improve on this, you can customize the order.

Both `\sym` and `\operatorsym` have an optional argument at the end. This defines the ordering. It is important to note that every symbol for which this last argument is given is placed behind every symbol without a last argument. Thus, it often only makes sense to give such an ordering argument either to all symbols or to none. As an example if “lca ” should be placed as lca in the ordering the definition should look like: `\operatorsym{lca}{The last common ancestor.}[lca]`.

Some times you want to group you symbols. This can be archived in two steps. First you must define the groups in the preamble. This is done with `\addSymbolCategory{C}{Cycle}` command. The first argument must be a *capital letter* and the second argument the name of the category. Second, first argument can then be used as last argument on the symbol definition to get a element in the category. It is important to note that this really only work with one capital letter. When more then one letter is given or a small letter LaTeX can not use the category. An example can be seen in Algorithm 4.

3.8.2 Symbols with Parameters

In many situations, you do not only have a symbol but also the symbols depend on other things. An example would be the vertex set of a graph. There are at least two different used notations for this $G.V$ or $V(G)$. However, it makes only sense when the graph (in this case G) is given. Thus a vertex set symbol needs a place that can be changed.

This behavior is reached with the `\macrosym` command. This command defines a macro that can be used in the thesis and also generates an entry in the list of symbols. Because this needs a name for the macro, its signature is different than the other methods.

Algorithm 5: The macro symbol command. Some examples of macro symbols and how they can be used.

```
%Let G be the 'graph' category.
\macrosym{V(#1)}{vertexset}[G]{The vertex set of $G$}[G]
\macrosym{E(#1)}{edgeset}[G]{The vertex edge of $G$}[G]
\macrosym{#1[#2]}{indsub}[G]{Creates induced subgraph of $G$
 with vertex set $U$ }[S][U]
%Let S be the 'set' category.
\macrosym{###1}{cardinality}[S]{The number of elements
in set $U$}[U]

When $D$ is a directed graph and $U\subset\symvertexset{D}$ and
$H=\symindsub{D}{U}$ have no cycles then is
$\symcardinality{\symedgeset{H}}\leq\frac{1}
{\symcardinality{U}(\symcardinality{U}-1)}$.
```

The command `\macrosym{content}{name}[C]{d}[p1][p2]...` creates the macro that is `\sym*name*`. The content is what is printed. It is given in latex macro form which means for the first parameter is "#1" and the second "#2" and so on. The category (C) and description (D) are the same as by the other commands. The command ends with a list of the parameters that are used in the index. There are up to 5 parameters possible. When you want a macro with four parameters you need to give four at the end of the command.

Algorithm 5 creates this output:

> When D is a directed graph and $U \subset V(D)$ and $H = D[U]$ have no cycles then is $\#E(H) \leq \frac{1}{\#U(\#U-1)}$.

3.9 Floats

There are three basic types of floats: figure, table and algorithm. Beside this, there are two special types: SCfigure and SCtable. These are the same as the floats without "SC". The difference is that in these floats the caption is placed beside the figure (table). This can be used with narrow figures (tables) to save space. An example of SCtable can be seen at Table 1.

3.9.1 Figures

The include `\includegraphics` command of a figure needs the path to the figure. The important part is that this path must be given relative to the main.tex file. For an example is the picture veelo.pdf in the Figures folder linked as "graphics/ell.pdf" in every Chapter.

3.9.2 Captions

The caption can be used with the normal `\caption` command. Remember, that the optional argument that sets the title on the "List of . . . " page. However, an

Algorithm 6: An example of how the `captiont` command can be used.

```
\begin{SCtable}
    \captiont{Font values.}%
    {The values that the class option \quo{font} can have.}%
    \label{tab:fontoption}
    \begin{tabular}{l r}
        \toprule
        \tblhead{Value} & \tblhead{Used Font}\\
        \midrule
        lm  & Latin modern without serifs (default)\\
        lms & Latin modern with serifs\\
        cm  & Computer modern without serifs\\
        cms & Computer modern with serifs\\
        \bottomrule
    \end{tabular}
\end{SCtable}
```

alternative in this class is the `\captiont` command that has a second mandatory argument that sets the title and also writes it as bold at the beginning of the caption. An example is shown in Algorithm 6.

All captions use a little smaller font than the rest of the text. This helps to distinguish between text and caption of a float. This supports the reading flow.

The caption is placed above or under the content of the float depending on the position of the caption command. However, normally the figure caption should be placed under the content, in a table above the content and in an algorithm also above the content.

3.10 Code

This template provides two different ways to typeset program code. One is meant for code written in a real programming language and includes syntax highlighting, the other is for pseudo-code that is more abstract than real code. These variants use the `code` and the `algorithmic` environments, respectively. However, in both cases, a surrounding *float*, which is automatically positioned and provides a caption that can be labelled and referenced, is useful. This feature is available in form of the `algorithm` environment, which then includes either a `code` or a `algorithmic` environment. An example is shown in Algorithm 7.

3.10.1 Pseudo Code

To typeset pseudo code, the `algorithmic` package is used. See Algorithm 8 for how the code looks like For more details what commands exist, please have a look at the manual.[1]

[1] `ftp://ftp.fu-berlin.de/tex/CTAN/macros/latex/contrib/algorithms/algorithms.pdf`

Algorithm 7: Example of an `algorithm` float enclosing a `code` environment to display its own code.

```
\begin{algorithm}
    \caption{%
        Example of an \cmd{algorithm} float enclosing a
        \cmd{code} environment to display its own code.
     }%
    \label{alg:algo}
    *code environment*
\end{algorithm}
```

Algorithm 8: The `algorithmic` environment can be used to typeset pseudo-code.

```
\begin{algorithmic}
    *pseudo-code*
\end{algorithmic}
```

Algorithm 9: An example how code highlighting works.

```
\begin{code}{python}
    *code*
\end{code}
```

3.10.2 Real Code with Highlighting

The `code` environment is used to typeset code written in a real programming language. Passing the name of a supported programming language, e. g. `\begin{code}{python}`, allows the package to apply syntax highlighting to the code. An example is given in Algorithm 9. Though `code` environments can be used by itself to place the code exactly at the given position, you should usually enclose it by an `algorithm` environment to allow automatic positioning and provide a proper caption as in Algorithm 7.

For code highlighting, two different packages are used. The default package is the `listings` package. This is nice because it works out of the box on any LaTeX distribution. To achieve nicer syntax highlighting (with color), another package is used: the `minted` package. It depends on the external *Python* program *Pygments*. In order to use it, you need to install them. More information on this can be found online.[2] The manual is created with `minted`.

minted To use `minted` instead of `listings` you set the class option `minted`.

However, which package you use does not have an influence on the environment name, which is always `code`. This makes it possible to switch between both options on the fly.

[2] `https://github.com/gpoore/minted`

3.11 Math Environments

There are six different environments that belong to this class. They all have their own uses. The explanations of them are taken from Richeson (2008).

3.11.1 Proposition

A proved and often interesting result, but generally less important than a theorem.

```
\begin{proposition}
    *text*
\end{proposition}
```

3.11.2 Definition

A precise and unambiguous description of the meaning of a mathematical term. It characterizes the meaning of a word by giving all the properties and only those properties that must be true.

```
\begin{definition}
    *text*
\end{definition}
```

3.11.3 Corollary

A result in which the (usually short) proof relies heavily on a given theorem (we often say that “this is a corollary of Theorem A”).

```
\begin{corollary}
    *text*
\end{corollary}
```

3.11.4 Theorem

A mathematical statement that is proved using rigorous mathematical reasoning. In a mathematical paper, the term theorem is often reserved for the most important results.

```
\begin{theorem}
    *text*
\end{theorem}
```

3.11.5 Lemma

A minor result whose sole purpose is to help in proving a theorem. It is a stepping stone on the path to proving a theorem.

```
\begin{lemma}
    *text*
\end{lemma}
```

3.11.6 Proof

Every Corollary, Theorem and Lemma must be proven. Thus, such an environment, a Proof environment should follow.

```
\begin{proof}
    *text*
\end{proof}
```

3.12 Todo Notes

This is an example of a todo note. Try changing the `todo` option of the class and see what happens.

Often, some task is left undone and has to be finished later. This should be made visible in the thesis. The `\todo{}` command is the solution to this problem. It creates a colored note containing the todo text. The exact behaviour can be controlled with the class option `todo`, which can be set to any of the three values *margin*, *plain*, and *off*. The *margin* setting creates orange boxes located in the margin, keeping the layout intact and separating text and tasks. Additionally, a list of all remaining tasks is generated and placed on the first page of the thesis. This is probably the most elegant solution, but relies on the external package `marginnote`. The *plain* option simply inserts the task as red text, enclosed in brackets and with a smaller font size than the main text. It has no external dependencies. The value *off*, finally, allows to temporarily disable the output of todo notes completely without deleting them. This is useful for creating drafts of the thesis that should not contain the note text.

CHAPTER 4

Back Matter

Contents

4.1 Appendix

There may be more than one appendix. Each appendix should be thematically consistent. As an example, you could have one appendix for extra tables and one for pseudocode. However, it directly depends on your topic and how you present it. Still, the appendix should not be entirely unlinked to the main text. A small mention like "more data in Appendix X" is sufficient to justify an appendix.

Every appendix starts with a `\chapter` command. This command behaves differently in the appendix than in the main text. The numbering uses letters instead of numbers. Furthermore, the mini Table of Contents (ToC) is not present here. Generally, the formatting of such an appendix can be done more freely than in the main text. For example, one appendix could entirely consist of tables without any text except for the captions.

4.2 Lists of X

There are four "List of X" entries in the back matter. The first two are abbreviations and symbols. More details on both can be found in Section 3.7 and Section 3.8.

The other two are figures and tables. Both are in some way standard in a thesis. However, the real usage of them is small. Do you know someone that ever looked for a specific table or figure in these lists? So these two lists are per default not present in the back matter. However, if you still want to include them in the thesis, use the class options `figlist` (for figures) and `tablist` (for tables) .

figlist
tablist

4.3 Bibliography

The bibliography is created from one BibTeX file. This file is given in the preamble by the command `\addglobalbib`. It can be changed to anything else. In fact, the bibliography is created with biblatex, which also accepts other input formats than BibTeX. For an overview look at Lehman et al. (2018).

The use of `biblatex` means that the bibliography is not created with *BibTeX* but with *Biber*. This tool should be contained in any reasonably modern LaTeX distribution. However, many editors do not use it as default so you may need to change the configuration there.

The bibliography displays only those entries that are cited at some point in the thesis (see Section 3.4). With respect to the base on publications (see Section 2.3.2), it may be required to have some works in the file that are not cited anywhere. Note that the bibliography is created for the main text and the appendix, so by default all citations must be contained in this single file. You can, however, include other bib files by adding more `\addglobalbib` commands to the main LaTeX file.

fullcitename

By default, the first names are shortened to the first letter. This behavior can be changed with the class command `fullcitename`. When you use this command, the result is that the full first names are shown in all citations. Note that in many cases the downloadable *BibTeX* files only contains the shorted first name. When some of the citations use shortened and some complete names, the result will be an inconsistent look. To prevent this, use the command only when all you BibTeX entries have full first names.

4.3.1 BibTex

Even though BibLaTeX is more powerful than BibTex, there may be situations where you want to use BibTex instead, e. g. if Biber is not available. For this case, the class option `bibtex` can be used. BibTex keeps more or less all features intact. The style of the bibliography changes a little and the overall look is less pleasant. The same holds for the cite commands. Thus, I clearly recommend to use BibLaTeX. If you still want to use BibTex, you must also adapt the latexmkrc file to properly configure latexmk for using BibTex. Be aware that author style and annotations also does not work with BibTex. **bibtex**

4.3.2 Author Style

The author style is the way how BibLaTeX shows the author name in the bibliography. This is set with the class option `bibauthorstyle`. There are basically two options: first the given name, then the family name (`gf`); or first the family name, then a comma, and then the given name (`fg`). **bibauthorstyle**

More natural is to have the given name shown first. However, this can be disturbing when the bibliography is sorted after the family names. Thus, your brain may have a hard time to figure out where in the sorting you are. A possible solution is a mixed version of the style. This third possibility is to have the first author in `fg` order and all other authors in `gf` order (`fggf`).

You can use any of the three bold strings to both authorstyle class options. The based on bibliography (`baseonauthorstyle`) uses the `gf` order by default because the ordering of the papers in this smaller list is not so important. The main bibliography (`bibauthorstyle`) uses the `fggf` order such that most authors can easily be read but the ordering of the bibliography can also easily be followed.

4.3.3 BibLaTeX Annotations

BibLaTeX has the possibility to add annotations to a bib entry. These annotations must then be interpreted from the class. This thesis template interprets three different author annotations. These are the `underlinebase`, `underlineall`, and `firstauthor`.

First, it is important to know how author annotations are added. This is done with an extra field, the `AUTHOR+an` field. Examples can be seen in Algorithm 10. The content of the field is a list of entries that address the list of authors. Thus, every item in the list starts with an author position (starting with 1). Then follows a "=" and the actual annotations, each separated by a ",". The list items are then separated with a ";".

As an example, this field:
`AUTHOR+an = {1=underlinebase,firstauthor;2=firstauthor}`
means that the first author is annotated with `underlinebase` and `firstauthor`. The second author is annotated with `firstauthor`. Not every position has to be annotated. Thus, the field

Algorithm 10: An example how to use author annotations.

```
@article{van2000art,
    title={The art of writing a scientific article},
    author={Van der Geer, J and Hanraads, JAJ and Lupton, RA},
    AUTHOR+an = {1=underlinebase,firstauthor;2=firstauthor},
    journal={J. Sci. Commun},
    volume={163},
    number={2},
    pages={51--59},
    year={2000},
}
@misc{web:promotion2,
    author = "{Fakultät für Mathematik und Informatik}",
    AUTHOR+an = {1=underlineall},
    title = "Erste Änderungssatzung zur Promotionsordnung Fakultät für
              Mathematik und Informatik",
    year = "2017",
    howpublished = "\url{http://db.uni-leipzig.de/bekanntmachung/
                          dokudownload.php?dok_id=5033}",
    note = "[Online; accessed 27-August-2018]",
}
```

`AUTHOR+an = {3=underlinebase}`
would also be valid.

Now to the meaning of annotations. The `underlinebase` and the `underlineall` annotations do more or less the same thing. They underline the author name. A convenient way to use this is to underline your own name in every paper. This makes it easy for the reader to recognize your work. The difference between these two commands is that `underlinebase` only underlines the name in the "based on" bibliography (see Section 2.3.2) and `underlineall` also underlines it in the main bibliography. You may choose the preferred style according to your taste. Underlining in the based on bibliography is always a good idea. In the main bibliography, the underlined references may be a little lost in all the other ones that are not from you. However, if you cite many papers that you have done some work for but the thesis is not based on them, it could help to show all the work you have done.

The `firstauthor` annotation is used to mark authors that share the first authorship of a paper. In the main bibliography, this gives no extra information that helps finding the work. However, in the based on bibliography, the situation is different. Maybe you share the first authorship of a paper, but you are not at the first position. Then you can use this to make clear that this work is really from you. Thus, the annotation only effects the based on bibliography. Furthermore, note that this annotation only makes sense when more than one author is marked. In papers where only one first author exists, the annotation shall not be used.

Algorithm 11: An example how a curriculum scientiae is created.

```
\begin{cscategory}{Working Experience}
    \csentry{since 10/2018}{Tutor}[
        Institute for Computer Science, Leipzig University
        \bull{} Lecture: Algorithms and Data Structures
    ]
\end{cscategory}
```

4.4 Curriculum Scientiae

The Promotionsordung (Mathematik und Informatik, 2013) requires you to hand in a curriculum scientiae with you work. This is a curriculum vitae with a focus on your scientific work. It is not stated how you hand it in but it makes sense to make it a part of your thesis. To make it an extra file, some extra commands are provided by the class.

The first one is the `cscategory` environment. This gives you the environment where you can put `\csentry` in. For every topic (Education, Working Experience, IT Knowledge and so on), a new `cscategory` environment is created. Thus, the title of the topic is given as an argument to the `\begin{cscategory}{title}` command. Internally, the environment creates a tabular environment such that a `cscategory` must be complete on one page. However, this can be composited by breaking one category manually in parts. The second part should have a "(continued)" (or "(ctd.)" for short) in the title.

The `\csentry{left}{right}[secondline]` command has as a first argument a text that is positioned left on the page. This could be a date of an employment or a subcategory. The second one is then the text that is placed right on the page. This is could be a job description or some skills. When one line is not enough to describe the entry, there is an optional third argument that creates a second line on the right.

In case you want to create sub-entries for an entry, such as a grade for an education or a project of an employment, this can be done with the command `\bull`. This command is used in the second or third argument of the `\csentry` command. An example of the complete example is given in Algorithm 11.

4.5 Selbstständigkeitserklärung (Statement of Authorship)

The last thing that is required by the Promotionsordung (Mathematik und Informatik, 2013) is that you sign a Statement of Authorship. This is the last page of your thesis. It basically says that you have done no plagiarism.

There are two lines. The first one is for the location and date where you signed it. The second one is for your signature.

Appendices

APPENDIX A

Class Options Overview

Class Option	Description	Reference
abstractpath	The path of the abstract file	Section 2.4
acknowledgmentpath	The path of the acknowledgment file	Section 2.6
baseonauthorstyle	Change the author style in based on bibliography	Section 4.3.2
bibauthorstyle	Change the author style in main bibliography	Section 4.3.2
bibversion	Create the bib version title page	Section 2.1
bibtex	Use BibTex instead of BibLaTeX	Section 4.3.1
chapterstyle	Change the appearance of the chapter title page	Section 1.6.3
definitionindex	Add a Definition Index	Section 3.6
figlist	Add a List of Figures	Section 4.2
font	Change the font	Section 1.6.2
fullcitename	Use full names in bibliography	Section 4.3
infopath	The path of the thesis information file	Section 2.1
maxdis	After how many pages the full version of a abbreviation is used	Section 3.7
minted	Use the minted package for code highlighting	Section 3.10.2

Class Option	Description	Reference
noglossary	Deactivate abbreviations commands	Section 3.7
nominitoc	Deactivate the mini toc	Section 3.2.1
nostatistics	Deactivate the statistics of the Bibliographic Description	Section 2.3.1
smallautoref	Change autoref typs to small letters	Section 3.3
smallmargins	Change to small margins	Section 1.6.1
tablist	Add a List of Tables	Section 4.2
todo	Aktivate todo notes	Section 3.12
zusammenpath	The path of the German abstract file	Section 2.5

APPENDIX B

LaTeX Commands in the Class

B.1 Preamble Commands

Command	Description	Reference
`\addglobalbib`	Add a bib file	Section 4.3
`\addSymbolCategory`	Add a Symbol category.	Section 3.8
`\documentclass`	Load the document class.	Section 1.5
`\setauthor`	The full name of the author.	Section 2.1
`\setauthortitle`	The academic degree if the author.	Section 2.1
`\setbase`	Set publications the thesis is based on.	Section 2.3.2
`\setbirthplace`	The birth place of the author.	Section 2.1
`\setdayofbirth`	The day of birth of the author.	Section 2.1
`\setdayofdefens`	The day where you defended your thesis.	Section 2.1
`\setdedication`	Set the dedication the thesis (if any).	Section 2.2
`\setfilingdate`	The date where you submit your thesis.	Section 2.1
`\setfirstreviewer`	The first reviewer of your thesis.	Section 2.1
`\setgrade`	The grade that you get on your thesis.	Section 2.1
`\setkeywords`	Set keywords for the content of your thesis.	Section 2.2
`\setpublist`	Set publications for the list of your scientific publications.	Section 2.3.2
`\setsecondreviewer`	The second reviewer of you thesis.	Section 2.1
`\setsubtitle`	The subtitle of the thesis.	Section 2.1
`\settitle`	The title of the thesis.	Section 2.1

B.2 Commands

Command	Description	Reference
\abb	Place a abbreviation	Section 3.7
\abbs	Place the plural of an abbreviation	Section 3.7
\autoref	Reference to a label and add corresponding type of the reference	Section 3.3
\beginappendix	Begin the appendix	Section 1.5.2
\beginmainmatter	Begin the main matter	Section 1.5.2
\bull	Add new bullet point to a cs entry	Section 4.4
\caption	Add a caption	Section 3.9.2
\captiont	Add a caption with a title	Section 3.9.2
\chapter	Start a new chapter	Section 3.2
\citet	Cite a bib entry in the text	Section 3.4
\citep	Cite a bib entry in the parentheses	Section 3.4
\csentry	Add a entry to the cs	Section 4.4
\defabb	Define a abbreviation	Section 3.7
\include	Include an other file	Section 1.5.2
\indexbf	Create index entry and bold text	Section 3.6
\indexdef	Create index entry and margin note	Section 3.6
\indexnomn	Create index entry	Section 3.6
\label	Create a label that can be referred to	Section 3.3
\macrosym	Creates a macro and add it as symbol	Section 3.8
\makefrontmatter	Creates the front matter of the thesis	Section 1.5.2
\makebackmatter	Creates the back matter of the thesis	Section 1.5.2
\marginnote	Add a margin note	Section 3.5
\operatorsym	Add a operator symbol	Section 3.8
\part	Start a new part	Section 3.1
\ref	Reference to a label	Section 3.3
\section	Start a new section	-
\subsection	Start a new subsection	-
\sym	Add a symbol	Section 3.8
\todo	Add a todo note	Section 3.12

B.3 Environments

Environment	Description	Reference
`algorithm`	A algorithm float	Section 3.10
`algorithmic`	A code pseudo code environment	Section 3.10
`code`	A code highlighting environment	Section 3.10
`corollary`	A corollary environment	Section 3.11
`cscategory`	A new category for cs	Section 4.4
`definition`	A definition environment	Section 3.11
`figure`	A figure	Section 3.9
`lemma`	A lemma environment	Section 3.11
`proof`	A proof environment	Section 3.11
`proposition`	A proposition environment	Section 3.11
`SCfigure`	A figure with caption on the side	Section 3.9
`SCtable`	A table with caption on the side	Section 3.9
`table`	A table	Section 3.9
`theorem`	A theorem environment	Section 3.11

APPENDIX C

Version History

Until v1.1.3 (inclusive) from 2019, development was carried out by Fabian Gärtner.

Version 2.0.0 to 2.0.2 was developed by Felix Kühnl in 2022/23.

Version	Date	Description
2.0.2	2023-03-02	Added university and faculty name to README.md, link to homepage
		Improved formatting of the Pflichtexemplar title page
		Fixed `xassoccnt` counter errors (wrong page count) and warnings
		Improved formatting of this change log, added date column
2.0.1	2022-08-29	Improve documentation of `make` targets
		Generate website intro text from `README.md` using `make html`
2.0.0	2022-08-29	New Makefile using *latexmk*, supporting continuous compilation
		Dedicated document class for stand-alone documents with consistent look
		Refined directory structure, dedicated data files
		Consequent usage of logical markup
		Symbols can now be used in math and text with a single command
		Improve the `lms` font setting, make it the default
		Correctly handle line breaks in the (sub-)title
		Fixed many many small glitches, typos and bugs
		Added comprehensive `.gitignore` file
		Numerous improvements to the manual content
		Added manual section about common typographic mistakes
		Better-looking tables using commands from `\booktabs` package
1.1.3	2019-05-16	Add `baseonauthorstyle` and `bibauthorstyle` class option
		Add *BibLaTeX* annotations for underlining and authorship
1.1.2	2019-05-14	Fix a bug with page numbers when using `smallmargins`
1.1.1	2019-05-13	Add `smallmargins` class option
		Extend definition index
		Add symbol macro index
1.1.0	2019-03-28	Rewrite symbol commands
		Add definition index
1.0.2	2019-03-26	Line breaks on bib data page

Version	Date	Description
		Change bibliography to shorten names
		Add class option to change bibliography to full names
1.0.1	2019-03-14	Remove superfluous space in `\abb` command
		Write text to different chapter titles
1.0.0	2019-01-16	Make it run able on bioinf computer
		Possible to deactivate statistics
		Possible to use *BibTeX*
		Possible to deactivate abbreviation commands
		Default all autorefs have a uppercase first letter
		Possible to set autorefs to lowercase
		Use `todonotes` package for `todo` command
		Corrections in the manual
		Bug fix in the reference counter
0.9.1	2018-10-22	Add missing `amsthm` package
0.9.0	2018-10-03	Add section for main file
		Add section for appendix
		Appendix A written
		Appendix B written
0.8.1	2018-09-15	Use `microtype` package
		Fix spacing on title page
0.8.0	2018-09-10	First published version

Please update this document as you commit changes to the code base.

List of Abbreviations

SV Directed Acyclic Graph.

SV minimum Feedback Arc Set.

SV Super Vertex.

SV Table of Contents.

SV Coordinated Universal Time.

Definition Index

Bibliography

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Curriculum Scientiae

Personal Information

Name	Fabian Gärtner
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Birthplace	Herdecke

Education

since 10/2014	PhD Student Bioinformatics Group Universität Leipzig
10/2011–09/2014	Master Student Computer Science focus on Bioinformatics at Universität Leipzig • Thesis: Oriented Graph Grammars and Stereochemistry • Grade: 1.9
10/2008–09/2011	Bachelor Student Computer Science at Universität Leipzig • Thesis: Vorhersage und Analyse von konservierten Merkmalen der Kontrollregion mitochondrialer Genome • Grade: 2.4

Working Experience

since 10/2014	Member of ScaDS Competence Center for Scalable Data Services and Solutions
since 10/2018	Tutor Informatics Universität Leipzig • Lecture: Algorithm and Data Structure
04/2016–09/2016	Tutor Informatics Universität Leipzig • Lecture: Algorithm and Data Structure • Lecture: Modeling and Programming

Working Experience (continued)

10/2012–03/2014	Teaching Assistant Informatics Universität Leipzig • Lecture: Algorithm and Data Structure
04/2012–09/2012	Teaching Assistant Informatics Universität Leipzig • Lecture: Practical Java Course
03/2009–03/2011	Research Assistant Bioinformatics Universität Leipzig • Project: MITOS: Improved de novo Metazoan Mitochondrial Genome Annotation

IT Skills

Operating systems:	MacOS, Linux, Windows
Programming:	Java, Python, C++, R, PHP, Javascript, Bash, Swift, MySQL
Markup languages:	Latex, HTML, CSS
Frameworks:	Flink, Tinkerpop, Hadoop

Languages

German:	native speaker
English:	fluent

Selbstständigkeitserklärung

Hiermit erkläre ich, die vorliegende Dissertation selbstständig und ohne unzulässige fremde Hilfe angefertigt zu haben. Ich habe keine anderen als die angeführten Quellen und Hilfsmittel benutzt und sämtliche Textstellen, die wörtlich oder sinngemäß aus veröffentlichten oder unveröffentlichten Schriften entnommen wurden, und alle Angaben, die auf mündlichen Auskünften beruhen, als solche kenntlich gemacht. Ebenfalls sind alle von anderen Personen bereitgestellten Materialien oder erbrachten Dienstleistungen als solche gekennzeichnet.

Ort, Datum

Unterschrift